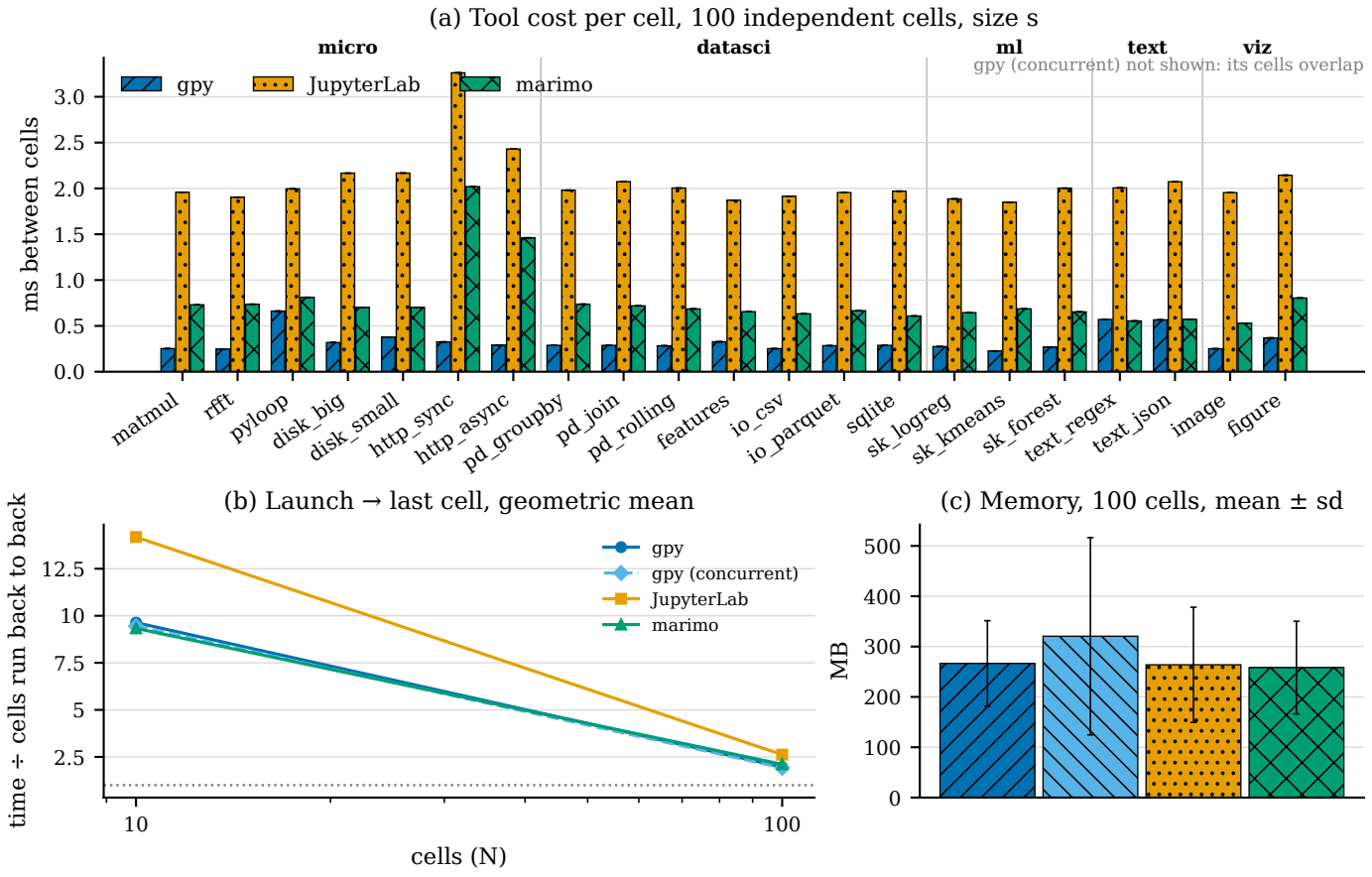

(a) Tool cost per cell, 100 independent cells, size s
micro
datasci
ml
text
viz
gpy
JupyterLab
marimo
gpy (concurrent) not shown: its cells overlap
ms between cells
0.0
0.5
1.0
1.5
2.0
2.5
3.0
matmul
rfft
pyloop
disk_big
disk_small
http_sync
http_async
pd_groupby
pd_join
pd_rolling
features
io_csv
io_parquet
sqlite
sk_logreg
sk_kmeans
sk_forest
text_regex
text_json
image
figure
(b) Launch → last cell, geometric mean
time ÷ cells run back to back
2.5
5
7.5
10
12.5
gpy
gpy (concurrent)
JupyterLab
marimo
10
100
cells (N)
(c) Memory, 100 cells, mean ± sd
MB
0
100
200
300
400
500